FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – I
PAPER –1
TEST DATE: 30-06-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

  Section – A *(01 – 04, 20 – 23, 39 – 42)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

  Section – A *(05 –10, 24 – 29, 43 – 48)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

  Section – B *(11 – 13, 30 – 32, 49 – 51)*: This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

  Section – C *(14 – 19, 33 – 38, 52 – 57)*: This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

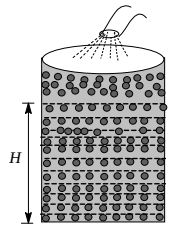
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A cylindrical container having radius r = 0.3 m has perforated wall. There is large number of uniformly spread small holes on the vertical wall occupying a fraction $\eta = 0.02$ of the entire area of the wall. To container, water is being fed to it at a constant rate $Q(m^3s^{-1})$ and at steady state its height becomes 2 m. Find Q.

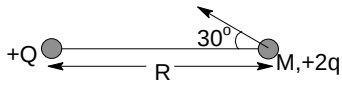


(A) 0.32
(B) 0.36
(C) 0.2
(D) 0.12

2. A ball is projected from a point 10 m away from a wall. If the initial velocity with which the ball can be projected is 15 m/s, then in order to hit the wall at the highest possible point the angle with the horizontal that it must be projected is
(A) 45
(B) 90
(C) $\tan^{-1} 2.25$
(D) $\tan^{-1} 1.5$

3. A light thread is wound four times around a pulley disc of diameter 30 cm and mass 5 kg. We want to unwind the thread by pulling it in 1 s. The minimum constant force required to do so is:
(A) 12.25 N
(B) 18.85 N
(C) 24.5 N
(D) 37.7 N

4. In the diagram shown, the charge +Q is fixed. Another charge $+2q$, is projected from a distance R from the fixed charge. The minimum separation between the two charges when the velocity becomes $1/\sqrt{3}$ times of the projected velocity, is (Assume gravity to be absent)

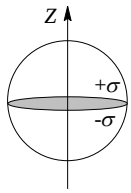


(A) $\frac{\sqrt{3}}{2}R$
(B) $\sqrt{3}R$
(C) $\frac{1}{2}R$
(D) $4R$

Section – A (Maximum Marks: 24)

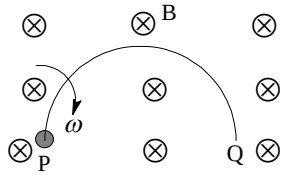
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A hollow, insulating spherical shell has a surface charge distribution placed upon it, such that the upper hemisphere has a uniform surface charge density $+\sigma$, while the lower hemisphere has a uniform charge density $-\sigma$, as shown in the figure. Their interface lies in x-y plane. Which of the following statement(s) is/are correct?



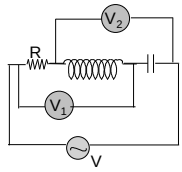
(A) The field at all points of x-y plane within the sphere points in the $-ve$ z-direction
(B) All points of the x-y plane within the sphere are equipotential
(C) The field at all points on z-axis outside the sphere point along positive z-direction
(D) The field at points on z axis which are on either side of origin outside the sphere is in opposite directions.

6. A wire is bent to form semi-circle of radius a. The wire rotates about its one end with angular velocity ω. Axis of rotation is perpendicular to plane of the semicircle. In the space, a uniform magnetic field of induction B exists along the axis of rotation as shown in figure. Then



(A) Potential difference between P and Q is equal to $2B\omega a^2$
(B) Potential difference between P and Q is equal to $2\pi^2 B\omega a^2$
(C) P is at higher potential Q
(D) P is at lower potential than Q

7. In an R-L-C series circuit shown in the figure, the readings of voltmeters V_1 and V_2 are 100V and 120V, respectively. The source voltage is 130V. For this situation, mark out the correct statement(s).



(A) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}$ V and $120+50\sqrt{3}\,\text{V}$, respectively
(B) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}$ V and $120-50\sqrt{3}\,\text{V}$, respectively
(C) Power factor of the circuit is $\frac{5}{13}$
(D) The circuit is capacitive in nature

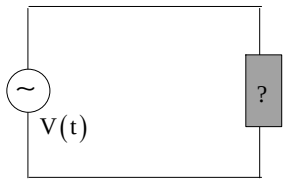
8. An α-particle of mass m_α and q_α moves a velocity $V_0=\left(2\hat{i}+3\hat{j}-\hat{k}\right)\text{ms}^{-1}$ through a region, where both a uniform magnetic field and a uniform electric field. Which of the following statements is/are correct? [Given, $B=\left(2i+4j+k\right)\text{T}$ and $E=\left(4i-j-2k\right)\text{VM}^{-1}$]

(A) Acceleration of α-particle is perpendicular to Z-axis
(B) Acceleration of α-particle is parallel to Z-axis
(C) de-Broglie's wavelength associated with α-particle is increasing with time
(D) de-Broglie's wavelength of α-particle keeps on decreasing with time

9. The polonium isotope $^{210}_{84}\mathrm{Po}$ is unstable, it emits an α-particle with energy of 5.30MeV. The atomic mass of $^{210}_{84}\mathrm{Po}$ is 209.9829 u and atomic mass of $^{4}_{2}\mathrm{He}$ is 4.0026 u. Then,

(A) Daughter nucleus is $^{206}_{82}\mathrm{Pb}$
(B) Disintegration energy of α-decay is 5.30MeV.
(C) Disintegration energy of α-decay is 5.40MeV.
(D) Mass of daughter nucleus is 205.9745u.

10. An AC voltage source is connected to a "black box" which contains a circuit, as shown in Figure. The elements in the circuit and their arrangement, however, are unknown. Measurements outside the black box provide the following information



V(t) = (80V) sin ωt
I(t) = (1.6A) sin (ωt+π/4)
Based on the above information we can infer:
(A) The circuit is largely capacitative
(B) The circuit in the black box is in resonance
(C) The box must contain at least one resistor and one capacitor
(D) The average power delivered to the black box by the AC source is 45.25 W

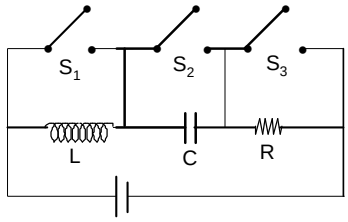
Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. A spherical planet of radius r, painted black, travels around the sun in a circular orbit at a distance D from its centre $(r<<<D)$. The sun, a sphere of radius R, radiates as a blackbody at a temperature $T_0=6000K$ and subtends an angle 0.01 rad as seen from the planet. The equilibrium temperature T of the planet is $x\times100$. find x

12. Nuclei of radioactive element A are produced at rate $'t^2'$ at any time t. The element A has decay constant λ. Let N be the number of nuclei of element A at anytime t. AT time $t=t_0, \frac{dN}{dt}$, is minimum. Then the number of nuclei of element A at time $t=t_0$ is found to be $\frac{Nt_0-\lambda t_0^2}{\lambda}$. Find the value of N

13. Consider the circuit shown in figure. With switch S1 closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch S2 closed and the other two switches open, the circuit has a time constant 2 sec. With switch S3 closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take pie2 = 10)



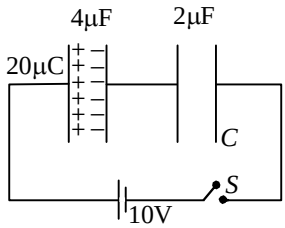
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

A $4\mu F$ capacitor is given $20\mu C$ charge and is connected with an uncharged capacitor of capacitance $2\mu F$ as shown in figure. When switch *S* is closed.



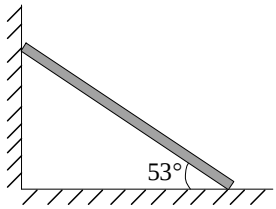
14. Charge flown through the battery is. (in μC)

15. Work done by the battery is (in μJ)

Question Stem for Question Nos. 16 and 17

Question Stem

A rod of length 10 meter has one end on smooth floor and the other end on smooth wall is released from the rest from the position shown in figure. When the rod makes an angle of 37° with the horizontal, answer the following:



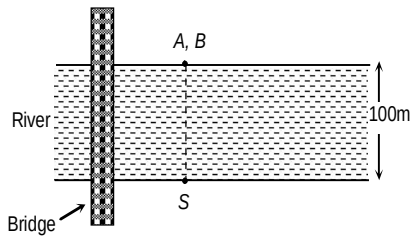
16. Square of the angular velocity of rod is: (in rod/sec)

17. The velocity of centre of mass of rod is: (in m/sec)

Question Stem for Question Nos. 18 and 19

Question Stem

There is a river of 100 m wide. Two persons *A* and *B* are located at same position. Person *A* can observe frequency greater than 200 Hz and person *B* less than 200 Hz. A sound source of frequency 200 Hz, which is located at point *S*, perpendicularly opposite to *A*, *B* location, at $t = 0$ is trying to cross the river perpendicularly with the velocity of 3 m/sec, where as velocity of river is 4 m/sec. The moment source reaches the other end of river, it stops and persons *B* starts moving towards the source whereas person *A* moves towards the bridge with equal uniform velocity.



18. For what duration person *B* observes the signal (in sec)

19. For what duration person *A* observes the signal (in sec)

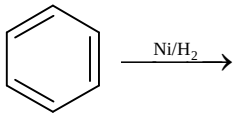
Chemistry

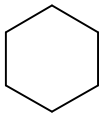
PART – II

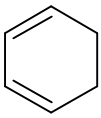
Section – A (Maximum Marks: 12)

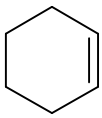
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

20. The product which can be best isolated from this reaction given below will be:



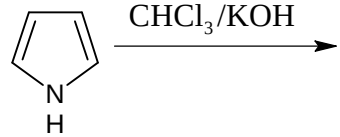
(A) 

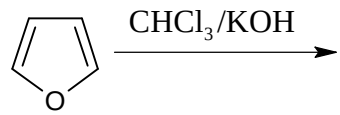
(B) 

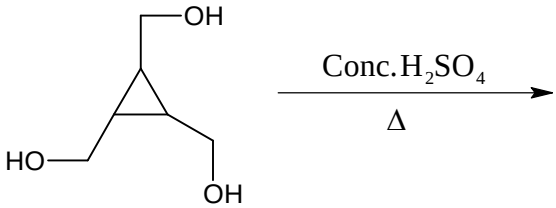
(C) 

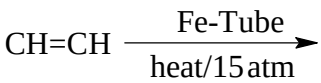
(D) Mixture of (A), (B) and (C)

21. Which among the following change cannot give six membered aromatic substance as one of the product.

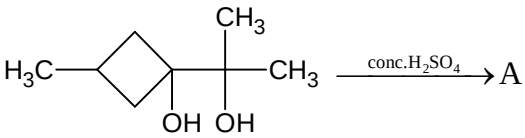
(A) 

(B) 

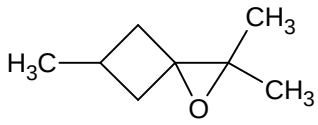
(C) 

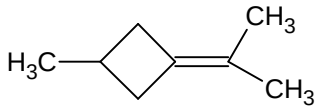
(D) 

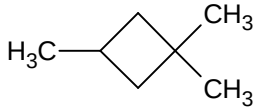
22. In the reaction

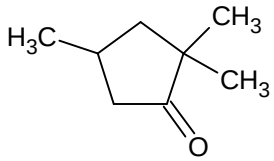


The product A is:

(A) 

(B) 

(C) 

(D) 

23. What is the percent by mass of titanium in rutile, a mineral that contain titanium and oxygen, if structure can be described as a closet array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium?

(A) $\approx 50\%, +4$

(B) $\simeq 60\%, +2$

(C) $\simeq 60\%, +4$

(D) $\simeq 30\%, +2$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

24. Which of following IUPSC name is/are nearly correct (ignoring alphabetical order) for the complex $\left[\mathrm{CO}\left(\mathrm{NH_3}\right)_2\left(\mathrm{tn}\right)\left(\sigma-\mathrm{C_3H_5}\right)_2\right]\mathrm{NO_3}$?

(A) Diallyldiamminetrimethylenediaminecobalt (III) nitrate
(B) 1, 3-Diaminopropanediamminediallylcobalt (III) nitrate
(C) Diammine-1, 3-diaminopropanedicyclopropyl cobalt (III) nitrate.
(D) Diallyldiammine-1, 3-diaminopropanecobalt (III) nitrate.

25. Which of the following on hydrolysis will give a product with specific rotation of $+52.7°$?

(A) Maltose
(B) Cellobiose
(C) Sucrose
(D) Cellulose

26. Identify incorrect statement.

(A) D-glucose, D-mannose and D-fructose all give same osazone on reaction with $\mathrm{Ph\text{-}NH\text{-}NH_2}$

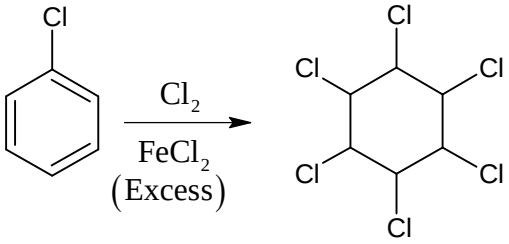
(B) D-glucose and D-fructose on catalytic reduction give same products.

(C) Glucose never react with $\mathrm{NaHSO_4}$ Schiff's and $\mathrm{Ph\text{-}NH\text{-}NH_2}$

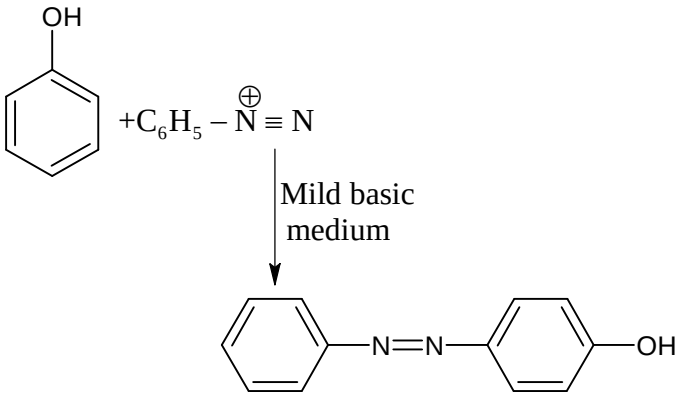
(D) Monomer of natural rubber is neoprene.

27. Which of the following reaction(s) is/are not representing the correct major product?

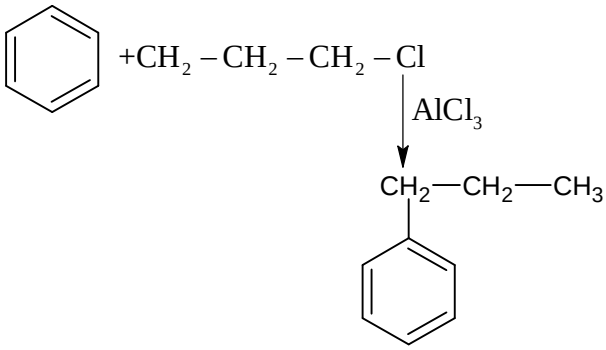
(A)



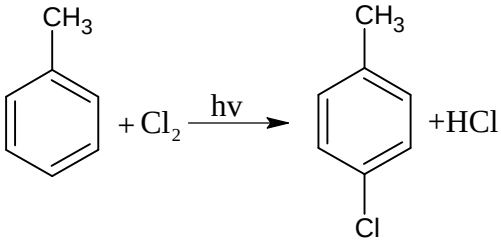
(B)



(C)

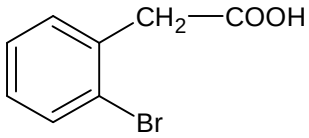


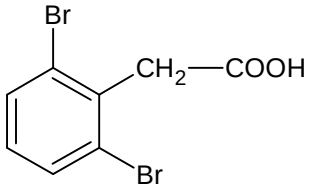
(D)

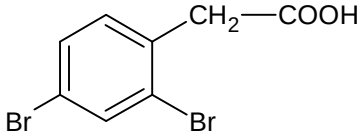


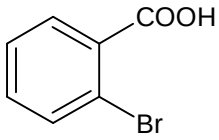
28. An acid (P), $C_8H_7O_2Br$ on bromination in the presence of $FeBr_3$ gives two isomers, (Q) and (R) of formula $C_8H_6O_2Br_2$. Vigorous oxidation of (P), (Q) and (R) gives acids (S), (T) and (U) respectively. (S), $C_7H_5O_2Br$ is the strongest acid among all of its isomers, whereas (T) and (U) each has a molecular formula of $C_7H_4O_2Br_2$.

(A)

P is 

(B) T is 

(C) U is 

(D) S is 

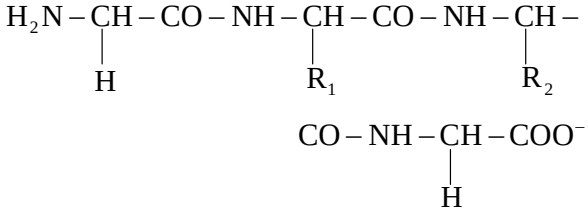
29. Sulphur dioxide can be qualitatively or quantitatively determined using:
(A) Starch iodate paper
(B) Starch iodide paper
(C) Oxidation to H_2SO_4
(D) $K_2[HgCl_4]$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. Consider following polymers:
Polythene, PVC, Bakelite, Nylon-6,6, Dacron, Buna-N, Buna-S
Neopren, Teflon, Metaldehyde
Number of homopolymer = X
Number of co-polymer = Y
Number of addition polymer = J
Number of condensation polymer = K
'XYJK' will be your answer.

31. The substituents R_1 and R_2 for nine peptides are listed in the table given below. How many of these peptides are positively charge at pH = 7.0?



Peptides	R_1	R_2
I	H	H
II	$CH_2CH_2NHCH_3$	CH_2
III	CH_2COOH	H
IV	CH_2CONH_2	$(CH_2)_4NH_2$
V	CH_2CONH_2	CH_2CONH_2
VI	$(CH_2)_4NH_2$	$(CH_2)_4NH_2$
VII	$CH_2(COOH)$	CH_2CONH_2
VIII	CH_2OH	$(CH_3)_4NH_2$
IX	$(CH_2)_4NH_2$	CH_3

32. Blood freezes at 272.44K and a solution of 3.0g of urea in 250g of water freezes at 272.63K. Calculate osmotic pressure (in atm) of blood at 310K. (Assume density of blood at 310K to be 1g cm^{-3} and normal freezing point of water to 273K) then what is 100 times that osmotic pressure rounded to two decimal places

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

1 mol of CCl_4 vapours at 77°C occupies a volume of 35.0 L. If van der Waals constants are $a = 20.39\,\mathrm{L}^2$ atm mol^{-2} and $b = 0.1383\,\mathrm{L\,mol}^{-1}$.

33. Find compressibility factor z under low pressure region.

34. Find compressibility factor z under high pressure region.

Question Stem for Question Nos. 35 and 36

Question Stem

1 mol of A in 1 litre vessel maintained at constant T shows the equilibrium.

$\mathrm{A(g)} \rightleftharpoons \mathrm{B(g)} + 2\mathrm{C(g)} \qquad \mathrm{K}_{C_1}$

$\mathrm{C(g)} \rightleftharpoons 2\mathrm{D(g)} + 3\mathrm{B(g)} \qquad \mathrm{K}_{C_2}$

If the equilibrium pressure is $\frac{13}{6}$ times of initial pressure and $[\mathrm{C}]_{\mathrm{eq}} = \frac{4}{9}[\mathrm{A}]_{\mathrm{eq}^+}$

35. Calculate K_{C_1} ____

36. Calculate K_{C_2} _____

Question Stem for Question Nos. 37 and 38

Question Stem

Electrode potential of an electrode depends upon concentration of ions and temperature. Given

K_{sp} *for AgCl is* $1.0\times10^{-10}, E^{-}_{Co^{2+}|Co^{3+}} = -1.82V$

$E^{-}_{Ag^{+}|Ag} = 0.799V, \qquad E^{-}_{Cr_2O_7^{2-}|Cr^{3+}} = +1.33V$

37. Calculate E for a silver/silver chloride electrode immersed in 1.00M KCl at 25°C

38. Calculate the magnitude of potential corresponding to the following cell. Given

$\mathrm{Pt}\left|Co^{2+}(2.0M), Co^{3+}(0.01M)\right\|\left|Cr^{3+}(0.5M), Cr_2O_7^{2-}(4.0M), H^{+}(1.5M)\right|Pt$

Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. Image of the curve $xy = 1$ in the curve $\left(x+\sqrt{x^2+1}\right)\left(y+\sqrt{y^2+1}\right)=1$ (on coordinate plane) is

(A) $xy = 1$

(B) $xy + 1 = 0$

(C) $xy = 0$

(D) $x^2 + y^2 = 1$

40. In a triangle ABC (with usual notations), if ex-radii r_1, r_2, r_3 are in H.P., then $\frac{a+c}{b}$ is-

(A) 1
(B) 2
(C) 3
(D) 4

41. If A is an invertible square matrix of order n such that $|\text{adj}(\text{adj } A)| = |A|^{(2n^2-7n+7)}$, then sum of possible values of n is-

(A) 3
(B) 4
(C) 5
(D) 6

42. Let $f(x)$ is an even function such that $f(x) + f(x-3) = x(x-3)+1$, then $\int_0^3 \frac{f(x)dx}{(x^2-3x+1)}$ is

(A) $-\frac{2}{3}$

(B) $-\frac{3}{2}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. Let S_n denotes the sum of first n term of an arithmetic progression whose first term is -4 and common difference is 1. If $V_n = 2S_{n+2} - 2S_{n+1} + S_n (n \in N)$, then

(A) $V_n = \frac{-9n^2 + 5n - 12}{2}, (n \in N)$

(B) $V_n = \frac{n^2 + 5n + 12}{2}, (n \in N)$

(C) Minimum value of V_n is -9

(D) Minimum value of V_n is $-\frac{73}{8}$

44. The value of x satisfying the equation $x^9 + \frac{9}{8}x^6 + \frac{27}{64}x^3 - x + \frac{219}{512} = 0$ is-

(A) $\frac{-1+\sqrt{13}}{4}$

(B) $-\frac{1}{2}$

(C) $\frac{-1-\sqrt{13}}{4}$

(D) more than two real and distinct values

45. If α, β, γ are roots of $x^3 + 2x^2 - 3x + 1 = 0$, then value of $\frac{\alpha\beta}{\alpha+\beta} + \frac{\alpha\gamma}{\alpha+\gamma} + \frac{\beta\gamma}{\beta+\gamma}$ is less than

(A) 2
(B) 3
(C) 4
(D) 5

46. The function $f(x)$ is defined by $f(x) = \frac{ax+b}{cx+d}$, where $a, b, c \text{ and } d$ are non-zero real number such that $f(19) = 19, f(97) = 97 \text{ and } f(f(x)) = x$ for all values except $-\frac{d}{c}$. If range of $f(x)$ is $R - \{\alpha\}$ (where R denotes real number set), than

(A) Number of proper divisors of α is 2
(B) Number of proper divisors of α is 3
(C) Sum of all divisors of α is 180
(D) Sum of all divisors of α is 90

47. It is given that there are two sets of real numbers $A \equiv \{a_1, a_2, a_3, ...a_{100}\}$ and $\{b_1, b_2, b_3, ...b_{50}\}$. If $f : A \to B$ is such that every element in B has an inverse image and $f(a_1) \le f(a_2) \le ..... \le f(a_{100})$, then the number of such mapping is-

(A) ${}^{100}C_{51}$

(B) ${}^{99}C_{50}$

(C) ${}^{100}C_{49}$

(D) ${}^{99}C_{49}$

48. The real numbers x_1, x_2, x_3 satisfying the equation $x^3 - x^2 + bx + c = 0$ are in A.P. Which of the following are necessary?

(A) $b \le \frac{1}{3}$

(B) $c \ge -\frac{1}{27}$

(C) $b \le -\frac{1}{3}$

(D) $c \ge \frac{1}{27}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. Let z_1, z_2 and z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 2$. If $|z_1 - z_2| = |z_1 - z_3|$ *and* $z_2 \ne z_3$. Then find the value of $|z_1 + z_2||z_1 + z_3|$.

50. If $\lim_{x \to 1} \frac{100}{1 - x^{100}} - \frac{50}{1 - x^{50}} = 5A$; find the value of A.

51. If $\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \frac{1}{3^i 3^j 3^k} = \frac{27 \times A}{208}$; where A is single digit natural, then find the value of A.

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $y = f(x)$ be a solution of the differential equation $\frac{dy}{dx} = -4x(y-2)$ with $f(0) = 2$ and $g(x) = \max.\left\{1 + \frac{x}{2} - \left[\frac{x}{2}\right], \sin x, |x-1|\right\}$, $h(x) = \ln x, h(x) = \min.\left\{x^2 - 3x + 2, 2 + \sin x\right\}$

[Note: $[y]$ denotes greatest integer function less than or equal to y.]

52. Area enclosed by the curves $y=f(x), y = g(x)$ and the y-axis, is:

53. Number of solutions of the equation $f(x) = k(x)$ in $[0, 2\pi]$ is:

Question Stem for Question Nos. 54 and 55

Question Stem

Let $g(x)$ be a non-constant twice differentiable function defined on R (the set of all real numbers) such that $y = g(x)$ is symmetric about the line $x = 2$ and $g(-2) = g'\left(\frac{1}{2}\right) = g'(1) = 0$.

54. The value of $g(6)$ equals:

55. The minimum number of roots of the equation $g''(x) = 0$ in the interval $(0, 4)$ equals:

Question Stem for Question Nos. 56 and 57

Question Stem

In a triangle AOB, R and Q are the points on the side OB and AB respectively such that 3OR = 2RB and 2AQ = 3QB. Let OQ and AR intersect at the point P (where O is origin).

56. It the point P divides OQ in the ratio of $\mu : 1$, then μ is:

57. If the ratio of area of quadrilateral PQBR and area of ΔOPA is $\frac{\alpha}{\beta}$ then $(\beta - \alpha)$ is (where α and β are coprime numbers):